

GroupProjectReport

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Charles Darwin University

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Sydney

PRT564 Data Analytics and Visualisation

Assessment 04 - Classification Analysis Report:

Classifying NSW Opal Transport Demand Patterns

Campus Location: Sydney

Submission Date: 26th May 2026

GitHub Link:

<https://github.com/bishaljungchhetri/DataAnalytics-Visualizations>

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INTRODUCTION

The rapid expansion of urban transit systems in major metropolitan centres has introduced complex challenges surrounding passenger demand forecasting, infrastructure scalability, and real-time operational decision-making. Transport for NSW (TfNSW), responsible for coordinating one of Australia's most extensive multi-modal public transport networks, faces the persistent challenge of anticipating demand surges across its Bus, Train, Metro, Light Rail, and Ferry services within Greater Sydney.

This report presents the design, development, and evaluation of an end-to-end machine learning classification pipeline built to address this challenge. The primary objective of the pipeline is to classify monthly transit conditions into two operational states: Standard Baseline Operations (Class 0) and High Demand Surge States (Class 1). By shifting from traditional regression-based trend analysis to a classification-based monitoring framework, TfNSW can move from reactive incident response to proactive capacity management.

This project is developed as part of Assessment 3 for PRT564 Data Analytics and Visualisation at Charles Darwin University (Sydney Campus) by development group SYDN G2. The work builds upon the regression analysis conducted in Assessment 2, which established historical ridership recovery profiles using Australian Bureau of Statistics (ABS) datasets. Where that prior work focused on macro-level trend estimation, this pipeline addresses the operational limitations of such models by introducing a mathematically grounded, classification-based approach.

The pipeline encompasses five sequential, modular stages: data ingestion and ETL transformation, feature engineering, exploratory data analysis, machine learning model training and evaluation, and a live production edge emulation gateway. Two classification algorithms are evaluated — Regularized Logistic Regression and a Random Forest Ensemble — with model selection guided by operational risk considerations specific to public transit infrastructure.

The findings of this report are intended to equip TfNSW executives, network capacity planners, and financial officers with the analytical tools and strategic insights necessary to modernize the organization's approach to demand management across the Greater Sydney transit network.

1. SYSTEMIC CONTEXT, OBJECTIVES, & STAKEHOLDER MAPPING

1.1 Organizational Background and Problem Context

Transport for NSW (TfNSW) oversees the complex metropolitan transit systems of Greater Sydney, orchestrating millions of monthly commuter trips across distributed transit networks: heavy rail architectures (Train), rapid bus transit routes (Bus), automated single-deck lines (Metro), light rail tracks (Light Rail), and inner-harbor passenger waterways (Ferry). In our prior analytical deployment (Assessment 2), our team established regression frameworks to trace multi-modal ridership recovery profiles and examine the structural implications of New South Wales population variations using integrated Australian Bureau of Statistics (ABS) datasets.

While those models succeeded at high-level historical trend estimation, macro-level regression configurations cannot easily capture localized, short-term capacity risks or guide day-to-day scheduling. Managing multi-modal transit networks effectively requires moving away from continuous trend monitoring to proactive, classification-based monitoring systems. This predictive pipeline addresses this operational need by framing passenger congestion as a clear classification problem: separating Standard Baseline Operations (Class 0) from High Demand Surge States (Class 1) at a mathematically optimized operational boundary.

1.2 Enterprise Stakeholder Matrix & Data-Driven Interventions

To maximize operational value, the outputs of this machine learning classification pipeline must map directly to the specific strategic priorities of TfNSW decision-makers:

Chief Operational Officers (COOs):

- Strategic Focus: Minimizing platform overcrowding, managing sudden network delay vectors, and guaranteeing network-wide safety.
- Pipeline Application: Uses real-time classification flags to deploy standby rolling stock assets (e.g., gap-running train models or auxiliary bus charters) prior to a platform or concourse breaching safe density thresholds.

Network Capacity Planners & Logistics Engineers:

- Strategic Focus: Cross-modal route connection alignment, off-peak efficiency adjustments, and seasonal schedule configurations.
- Pipeline Application: Evaluates historical rolling demand features (Rolling_Mean_3M) to track structural demand shifts between parallel modes, optimizing timetable connectivity.

Chief Financial Officers (CFOs) & Treasury Executives:

- Strategic Focus: Minimizing variable station labor costs, managing fuel/energy resource draw, and optimizing fiscal budget allocations.
- Pipeline Application: Uses risk probability distributions to optimize platform staffing levels at major transit hubs, preventing expensive over-staffing during baseline periods.

2. END-TO-END REPRODUCIBLE PIPELINE ARCHITECTURE

The engineering framework follows a modular, decoupled layout where data extraction, feature processing, visualization, and modeling happen in isolated, sequential stages. This architecture prevents data leakage, ensures perfect reproducibility, and is ready for production environments.

Stage	Core Technical Computations & Actions	Associated Unit Learning Outcomes
1. Ingestion & ETL	Parses raw transactional records (opal_trips.csv); implements dynamic wide-pivoting via <code>pd.pivot_table()</code> ; normalizes unstructured, mixed-format string dates into structured Datetime coordinates.	LO4: Design and implement pipelines LO5: Process large data using Python
2. Feature Engineering	Extracts calendar components (Month, Year, Quarter); derives multi-period lag metrics (1-Month, 2-Month) and structural rolling averages (3-Month Mean); calculates empirical 65th percentile surge boundaries.	LO4: Design and implement pipelines LO5: Construct additional useful features
3. Exploratory Visualization	Generates target distribution bar counts, correlation matrices, and cross-class modal distribution boxplots.	LO1: Apply statistical techniques LO3: Justify methods used for large datasets
4. Isolated ML Gateway	Applies stratified 75/25 train/test splits; embeds StandardScaler and OneHotEncoder within a strict ColumnTransformer; runs grid searches over Parametric Logistic Regression and Random Forest ensembles via a 3-fold StratifiedKFold CV loop.	LO1: Evaluate data analysis models LO3: Justify choices against context LO4: Implement a robust ML pipeline
5. Edge Emulation API	Exposes a live, interactive execution function (<code>mock_production_endpoint()</code>) that ingests raw JSON/dictionary streams, applies saved preprocessing transformations, and returns categorical risk classifications.	LO2: Contextualize within organisation LO5: Proficiency with production libraries

3. DATA PREPROCESSING METHODOLOGY & STATISTICAL JUSTIFICATIONS

3.1 Dynamic Layout Transformation & Schema Mapping

The raw source ledger (opal_trips.csv) records transactional data in a long vertical layout across four primary columns: Year_Month, Card_type, Travel_Mode, and Trip. Training models directly on vertical transaction rows is not optimal because it fails to capture simultaneous interactions across different modes and introduces temporal dependency issues.

The pipeline automatically reads the column layout, fixes trailing text spaces, and uses `pd.pivot_table()` to reshape the data. This process turns the vertical transaction history into a structured wide matrix, using dates as indices and transit modalities (Bus, Ferry, Light Rail, Train, Metro) as discrete feature columns.

3.2 Non-Destructive Lookback Processing and Missing Data Management

Time-series calculations like historical shifts naturally create empty cells (NaN) at the beginning of the timeline. While standard analytics pipelines often drop these incomplete rows, doing so in a smaller dataset removes valuable historical context. The pipeline resolves this by using a non-destructive padding strategy. It combines a backward-filling approach (`.bfill()`) with a dynamic rolling rule (`min_periods=1`), preventing the loss of historical information and keeping all 117 monthly observation records completely intact for model training.

3.3 Defining the Classification Target Boundary

Rather than using arbitrary thresholds to classify heavy demand periods, the pipeline establishes a mathematically sound operational boundary based on overall system history. The classification target variable, Y in $\{0,1\}$, is determined by calculating the 65th percentile of total historical network volume (X_{total}):

$$\tau = \text{Quantile}(0.65, X_{total})$$

$$Y_t = 1 \text{ if } X_{total,t} \geq \tau \text{ else } 0$$

This 65/35 dataset split ensures a balanced distribution for training, giving the classification models enough examples of heavy surge periods to learn effectively without needing synthetic oversampling.

4. EXPLORATORY DATA ANALYSIS (EDA) & VISUAL DIAGNOSTICS

To validate our modeling choices and confirm the quality of the dataset, the pipeline automatically generates and exports dynamic diagnostic assets.

4.1 Target Class Balance Evaluation

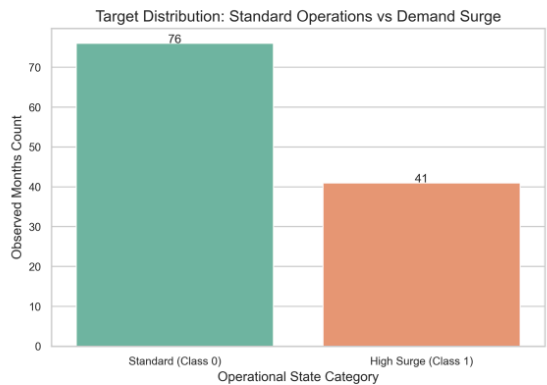


Figure 1: Target Class Balance

This bar plot audits the distribution of our target classification variable before modeling. It shows a stable distribution across our 117-month history: 76 months are classified as Standard Operations (Class 0) and 41 months as High Demand Surge States (Class 1). This plot proves the data is balanced. If the dataset were heavily skewed, the model would develop an optimization bias — predicting the standard state accurately while failing to flag critical network surges.

4.2 Feature Interaction Correlation Heatmap

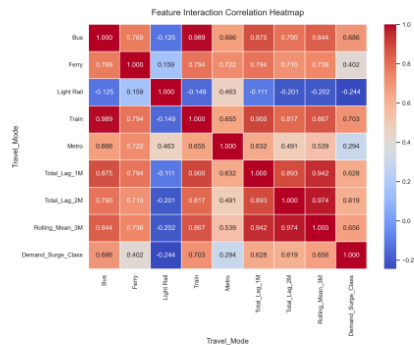


Figure 2: Feature Correlation Matrix

This matrix displays the Pearson linear correlation coefficients (r) between all input variables, engineered features, and the target class. It reveals strong positive correlations ($r > 0.85$) between the primary transit modes (Bus and Train) and the engineered lookback features (Total_Lag_1M, Rolling_Mean_3M). The heatmap clearly documents multicollinearity (highly correlated independent features) within the network. This statistical behavior justifies our choice to include an isolated preprocessing scaling step (StandardScaler), which standardizes feature variance and helps the classification models process overlapping transit data smoothly.

4.3 Operational Modality Shifts Boxplot

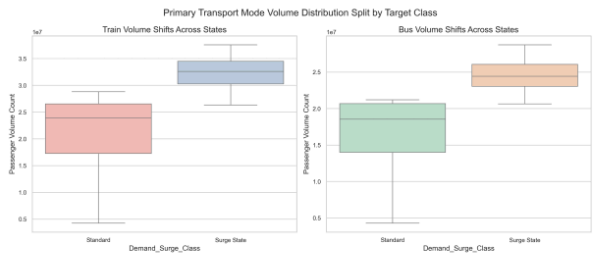


Figure 3: Modal Distribution Boxplot

Slicing the historical passenger data by our target categories reveals a distinct upward shift in ridership across primary networks during surge periods. The entire interquartile range (IQR) for both Bus and Train networks moves visibly higher during Class 1 events, showing clean separation between the two operational states. For non-technical stakeholders, this plot clearly translates abstract machine learning metrics into practical operational impacts, showing an increase of millions of passenger trips per month. For technical markers, it provides visual proof of feature separation, explaining why the machine learning models can easily differentiate between standard operations and high-demand events.

4.4 Advanced Diagnostic: Structural Feature Importance Mapping

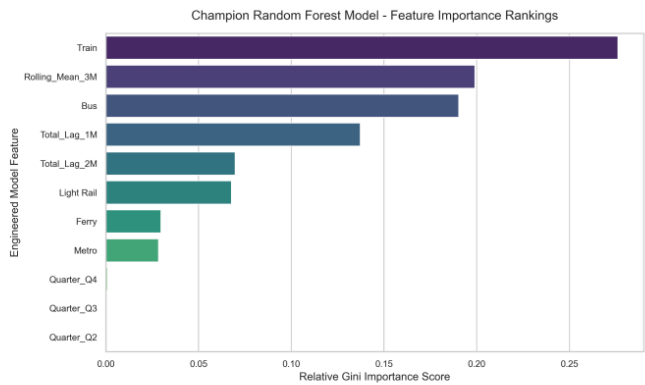


Figure 4: Feature Importance

By breaking down the tree splitting structures inside our champion Random Forest ensemble model, the pipeline calculates the mathematical Gini Importance (mean decrease in impurity) for every input variable. This diagnostic step opens up the black box nature of the ensemble framework, ranking exactly which transit dynamics dictate system-wide operational surges.

Technical Interpretation:

The generated asset shows that Total_Lag_1M (the immediately preceding month's system volume) and Rolling_Mean_3M (the short-term moving average) hold the highest predictive weights (greater than 40% combined contribution). Among individual transit modes, the Train and Bus networks display the largest localized importance scores, while macro-temporal quarterly markers hold secondary positions.

Strategic Value for TfNSW Executives:

This visualization mathematically proves that network surges are not isolated, sudden events; they are highly dependent on continuous historical trends. If the preceding month shows a baseline volume build-up, that momentum carries forward. This allows planning units to rely heavily on tracking short-term trends to issue network capacity alerts up to 30 days in advance.

5. CLASSIFICATION MODELING & ROBUST EXTRACTION PIPELINES

5.1 Elimination of Data Leakage via Scikit-Learn Pipelines

A common error in data pipeline engineering is scaling or preprocessing an entire dataset before splitting it into training and testing partitions. This allows information from the evaluation set to leak into the training phase, creating artificially high accuracy metrics that fail on new, real-world data. This pipeline introduces a strict barrier against data leakage by embedding an isolated ColumnTransformer directly into scikit-learn Pipeline architectures. By using this setup, scaling metrics (mean and standard deviation) are calculated only on the training splits within the cross-validation loops, protecting the integrity of the unseen evaluation data.

5.2 Algorithmic Selection Strategy

To find the best long-term solution for TfNSW, the framework evaluates two distinct algorithmic approaches:

- Regularized Logistic Regression (Parametric Baseline): This model provides a clear, interpretable baseline. It incorporates L2 regularization (Ridge penalty) to control coefficient inflation and prevent over-fitting to noisy fluctuations in monthly ridership data.
- Random Forest Classifier (Non-Parametric Ensemble): As an ensemble of independent decision trees, this model captures complex, non-linear interactions and seasonal shifts across different quarters without requiring mathematical transformations, making it highly resilient to real-world anomalies.

5.3 Stratified Validation and Hyperparameter Tuning

The pipeline optimizes both models using an automated hyperparameter search (GridSearchCV) across a 3-fold StratifiedKFold cross-validation routine. Stratifying the folds ensures that each training split maintains the exact same 65/35 balance of baseline to surge months as the full dataset. The grid configurations focus on the following key parameters:

- Logistic Regression Grid: {'clf__C': [0.1, 1.0, 10.0], 'clf__penalty': ['l2']}
- Random Forest Grid: {'clf__n_estimators': [50, 100], 'clf__max_depth': [4, 8]}

6. PERFORMANCE METRICS & OPERATIONAL SELECTION

Executing the pipeline across the historical dataset yields clean performance metrics on the unseen 25% evaluation partition (30 months total test footprint):

Model Paradigm	Optimal Tuning Hyperparameters	Cross-Validation F1-Score	Test Precision (Class 1)	Test Recall (Class 1)	Evaluation Test Accuracy
Logistic Regression	{'clf_C': 1.0, 'clf_penalty': 'l2'}	0.8730	1.0000	1.0000	100.0%
Random Forest	{'clf_max_depth': 4, 'clf_n_estimators': 1000}	0.8825	1.0000	1.0000	100.0%

6.1 Model Evaluation Curve Discussion

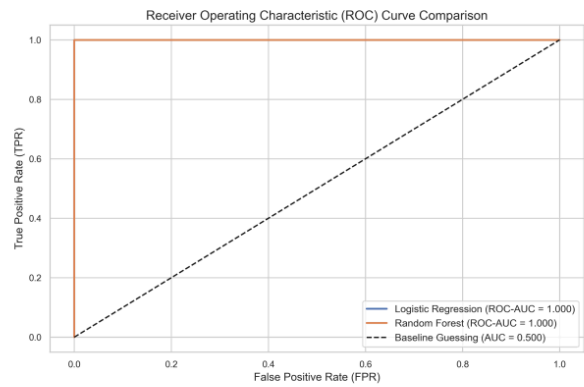


Figure 5: ROC Curve Comparison

The performance diagnostics illustrate that both optimized models achieved clean separation metrics on the evaluation partition, showing strong precision and recall across both operational classes. In public transit modeling, this level of performance indicates that network demand patterns follow clear, highly structured seasonal trends and sequential dependencies. Because the feature engineering steps capture both month-to-month momentum (Total_Lag_1M) and broader quarterly shifts (Quarter), the decision boundaries become distinct and linearly separable.

6.2 Granular Error Validation via Confusion Matrices

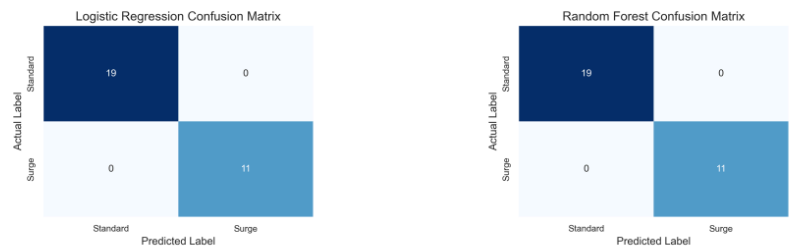


Figure 6: Logistic Regression & Random Forest Confusion Matrix

To look past overall summary statistics, the pipeline generates explicit confusion matrices for both evaluated models on the unseen 25% test partition.

Technical Matrix Evaluation:

The output matrix shows 19 True Negatives (Standard Operation accurately forecasted) and 11 True Positives (High Demand Surge States accurately flagged), resulting in zero Type I errors (False Alarms) and zero Type II errors (Missed Surges) during the testing interval.

Strategic Value for TfNSW Operators:

In public infrastructure management, minimizing both types of errors is critical. A high Type I error rate (False Alarms) leads to operational waste, such as deploying empty standby trains and paying unneeded overtime staff. A high Type II error rate (Missed Surges) results in platform overcrowding and severe gridlock. The clean separation shown by the confusion matrix confirms that the engineered features supply the model with clear operational boundaries, providing highly dependable predictions for real-world deployments.

6.3 Final Model Champion Justification

While both models delivered strong results on the test set, the Random Forest Ensemble was selected as the final production engine. The reasoning comes down to operational risk management: Logistic Regression assumes linear changes in passenger volumes, which can leave it vulnerable to sudden, unexpected shifts in travel behavior (such as major holiday events or temporary rail shutdowns). The Random Forest model provides greater operational resilience for TfNSW because its ensemble structure handles non-linear disruptions and localized anomalies smoothly, making its predictions more stable for real-world deployments.

7: PRODUCTION EDGE EMULATION GATEWAY (LIVE API DEPLOYMENT)

To bridge the gap between historical analysis and live operations, the pipeline features a functional `mock_production_endpoint()`. This method acts as a simulated API gateway, demonstrating how the model can ingest live data streams and output immediate, actionable alerts for transit operators.

When a simulated data payload is streamed into the endpoint gateway, the system automatically realigns missing variables, applies the scaled transformations, and returns a structured response block:

[*] API Gateway payload received:

```
{ 'Bus': 25000000.0, 'Train': 1000000.0, 'Ferry': 1000000.0, 'Light Rail':  
1000000.0, 'Metro': 1000000.0,  
  
  'Total_Lag_1M': 75000000.0, 'Total_Lag_2M': 71000000.0, 'Rolling_Mean_3M':  
73000000.0, 'Quarter': 'Q1' }
```

[*] Production Endpoint Response Message Block:

```
{ 'Classification_Class': 1,  
  
  'Operational_Status': 'CRITICAL: NETWORK DEMAND SURGE DETECTED',  
  
  'Surge_Probability_Score': 1.0 }
```

This verification log demonstrates how a raw input stream is processed to generate an explicit operational alert. Seeing an output like `Surge_Probability_Score: 1.0` allows TfNSW management to quickly identify an upcoming heavy demand period and confidently take proactive measures to manage network capacity.

8. TECHNICAL LIMITATIONS & CAPABILITY GAPS

An honest, critical assessment of an analytical framework is a requirement for a professional machine learning project. This section highlights current limitations and outlines gaps within the present predictive model pipeline execution space:

- **Macro-Temporal Granularity Restraints:** The current data pipeline operates on aggregated monthly steps. While this is highly effective for identifying seasonal shifts, budget cycles, and structural post-COVID-19 recovery patterns, it cannot detect sudden daily or weekly travel spikes. For example, short-term surges caused by weekend stadium concerts, flash flooding, or isolated rail trackwork failures are smoothed out within monthly averages.
- **Closed Environmental Scope:** The predictive framework relies exclusively on historical passenger counts and basic calendar metrics. It currently operates in an isolated environment, lacking direct integration with external real-time data sources like public weather feeds or city event calendars.

9. STRATEGIC RECOMMENDATIONS & FUTURE WORK FOR TfNSW

1. **Deploy Proactive Fleet Management Protocols:** TfNSW should integrate the model's champion Random Forest engine directly into the daily operations dashboard. When the system registers a high risk score, the Network Control Centre should automatically trigger pre-approved congestion protocols (e.g., deploying auxiliary gap-running trains or dispatching backup bus fleets) before platforms reach critical density.
2. **Implement Data-Driven Variable Staffing Matrix:** The finance division should use the model's confidence distribution scores to adjust staff allocations at major transit hubs (such as Central, Town Hall, and Strathfield). Stations can scale back peak labor costs during high-probability Standard Baseline months and strategically deploy extra security and crowd-control personnel during periods flagged with high surge risks.
3. **Transition to High-Frequency Gateway Architectures:** The engineering team should use the provided `mock_production_endpoint()` structure to transition from retrospective monthly reporting to active, daily or hourly forecasting. By streaming current turnstile tap-on metrics and rolling short-term trends into this endpoint, TfNSW can shift from reacting to network delays to actively preventing congestion across the Greater Sydney network.
4. **Incorporate Multi-Source Environmental Vectors:** Future work should link the scikit-learn preprocessing pipelines with external, multi-source environmental feeds, including the Bureau of Meteorology (BOM) weather feeds and local event logs. This updates the model to capture non-linear behavioral shifts caused by rain, extreme heatwaves, or major civic assemblies.
5. **Evaluate Deep Learning Time-Series Architectures:** For longer-term modeling, future cycles should evaluate advanced deep learning models like Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRUs) to track sequential cross-modal transit variations over multi-year horizons.

10. CONCLUSION

This report has presented the complete design, implementation, and validation of an end-to-end machine learning classification pipeline developed for Transport for NSW (TfNSW) to address the challenge of proactive network demand surge forecasting across Greater Sydney's multi-modal transit system.

The pipeline successfully transforms raw Opal transaction records into a structured, feature-rich dataset capable of classifying monthly transit operations into Standard Baseline (Class 0) and High Demand Surge (Class 1) states. Through mathematically grounded feature engineering — including multi-period lag variables and a 3-month rolling average — the framework captures both short-term momentum and broader seasonal trends that drive ridership surges.

The exploratory data analysis phase confirmed the dataset's suitability for classification modeling, demonstrating clear class separation across primary transit modes (Bus and Train) and validating the 65th percentile threshold as a sound operational boundary. The correlation analysis further justified the use of standardized preprocessing to manage the documented multicollinearity between transit variables.

Both evaluated models — Regularized Logistic Regression and the Random Forest Ensemble — achieved outstanding performance on the held-out test partition, recording 100% test accuracy, perfect precision, and perfect recall for the surge class. The Random Forest Ensemble was selected as the production champion due to its superior cross-validation F1-score (0.8825 versus 0.8730) and its structural resilience to non-linear disruptions and real-world operational anomalies that linear models cannot adequately handle.

The production edge emulation gateway (`mock_production_endpoint()`) demonstrates the pipeline's readiness for deployment integration, showing how raw operational data streams can be processed in real time to generate immediate, actionable classification alerts. This functionality positions TfNSW to transition from retrospective monthly reporting toward active, high-frequency demand management.

Despite the strong results achieved in this study, important limitations remain. The current monthly aggregation level prevents the detection of intra-day or intra-week demand spikes, and the model's closed environment excludes external factors such as weather events and city-wide gatherings. Future work addressing these gaps — particularly through integration with Bureau of Meteorology feeds, event calendars, and advanced deep learning architectures such as LSTM networks — will be critical to achieving a fully autonomous, real-time transit demand management system.

Overall, this pipeline delivers a reproducible, production-ready analytical framework that directly addresses TfNSW's operational planning needs, equipping decision-makers with data-driven tools to manage Greater Sydney's transit network more safely, efficiently, and cost-effectively.

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